

A Separable-Range Counterexample to the Injective-Ideal ℓ_∞ -Extension Question

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Source and Question

The source paper is A. K. Karn and D. P. Sinha, *Compactness and an approximation property related to an operator ideal*, arXiv:1207.1947. In Section 4 the authors define the ℓ_∞ -extension property for an operator ideal $(\mathcal{A}, \alpha)$: whenever

$$T \in \mathcal{A}(X, \ell_\infty(\Gamma)),$$

there should exist

$$\tilde{T} \in \mathcal{A}(\ell_\infty(B_{X^*}), \ell_\infty(\Gamma))$$

such that $T = \tilde{T}i_X$ and $\alpha(\tilde{T}) = \alpha(T)$, where $i_X : X \rightarrow \ell_\infty(B_{X^*})$ is the Alouglu embedding.

Near the end of the paper, after Corollary 6.6, the authors write:

“we do not know whether there is an injective operator ideal that fails the ℓ_∞ -extension property.”

spaces Y . Thus X^* has the $\mathcal{A}$ -a.p. if and only if $((\mathcal{A}^{uuu})^{uuu})^{uuu}(X, Y) = (\mathcal{A}^{dual})^{min}(X, Y)$, for all Banach spaces Y . $\triangle$

Again note that since Π_p is injective and has the ℓ_∞ -extension property, Theorems 4.5 and 4.6 in [8] are special cases of the above corollary. Let us also note here that we do not know whether there is an injective operator ideal that fails the ℓ_∞ -extension property.

Corollary 6.7 *Let $\mathcal{A}$ be a right accessible surjective operator ideal and X be a Banach space. Then we have the following:*

(a) *X has the $\mathcal{A}$ -approximation property if and only if*

Result

The answer is affirmative. The operator ideal of separable-range operators is injective, but it fails the ℓ_∞ -extension property.

Definitions

Let $\mathfrak{X}(E, F)$ denote the class of bounded linear operators $T : E \rightarrow F$ for which $\overline{T(E)}$ is separable, equipped with the ordinary operator norm. This is a closed Banach operator ideal: sums and ideal compositions preserve separability of the closed range, and norm limits have range contained in the closed linear span of countably many separable ranges.

Proof Intuition

The ℓ_∞ -extension property would extend every separable-range map into an ℓ_∞ space while keeping the extension's range separable. Apply this to the canonical embedding of c_0 into its Alouglu envelope. Any separable-range extension has a separable closed range containing the copy of c_0 . Sobczyk's theorem then complements that copy of c_0 inside the separable range. Composing the projection with the hypothetical extension gives a projection from the Alouglu envelope onto c_0 . But a Banach space complemented in its Alouglu envelope is injective; this would force c_0 to be injective, contradicting Phillips' theorem.

Lemma 1. *The ideal $\mathfrak{X}$ of separable-range operators is injective.*

Proof. Let $J : F \rightarrow G$ be an isomorphic isometry and let $T : E \rightarrow F$ be bounded. If T has separable range, then JT has separable range. Conversely, if JT has separable range, then $J(T(E))$ is separable, and applying J^{-1} on the subspace $J(F)$ shows that $T(E)$ is separable. Thus $T \in \mathfrak{X}(E, F)$ if and only if $JT \in \mathfrak{X}(E, G)$, which is precisely injectivity of the ideal. $\square$

Theorem 1. *The injective Banach operator ideal $\mathfrak{X}$ does not have the ℓ_∞ -extension property.*

Proof. Let $X = c_0$ and identify $X^* = \ell_1$. Write

$$i = i_X : c_0 \longrightarrow \ell_\infty(B_{\ell_1})$$

for the Alouglu embedding. Since c_0 is separable, i has separable range, so

$$i \in \mathfrak{X}(c_0, \ell_\infty(B_{\ell_1})).$$

Suppose, toward a contradiction, that $\mathfrak{X}$ had the ℓ_∞ -extension property. Then there would be an operator

$$S \in \mathfrak{X}(\ell_\infty(B_{\ell_1}), \ell_\infty(B_{\ell_1}))$$

such that

$$Si = i.$$

The range of S is separable; let

$$E = \overline{S(\ell_\infty(B_{\ell_1}))}.$$

Then E is a separable Banach subspace of $\ell_\infty(B_{\ell_1})$, and $i(c_0) \subset E$ because $Si = i$.

By Sobczyk's theorem, every copy of c_0 inside a separable Banach space is complemented. Hence there is a bounded projection

$$P : E \rightarrow i(c_0).$$

Now PS , viewed as an operator from $\ell_\infty(B_{\ell_1})$ into $\ell_\infty(B_{\ell_1})$, is a bounded projection onto $i(c_0)$: its range is contained in $i(c_0)$, and for every $x \in c_0$,

$$PS(ix) = P(ix) = ix.$$

Thus $i(c_0)$ is complemented in its Alouglu envelope $\ell_\infty(B_{\ell_1})$.

We next recall the standard consequence. If a Banach space Y is complemented in $\ell_\infty(B_{Y^*})$ through its Alouglu embedding, then Y is injective. Indeed, for any isometric embedding $Y \subset Z$, injectivity of the ℓ_∞ space extends the Alouglu embedding $i_Y : Y \rightarrow \ell_\infty(B_{Y^*})$ to a bounded operator $R : Z \rightarrow \ell_\infty(B_{Y^*})$; composing R with the projection onto $i_Y(Y)$ and then with i_Y^{-1} extends the identity on Y to a projection from Z onto Y .

Therefore the preceding paragraph would make c_0 injective. This is impossible: by Phillips’ theorem, c_0 is not complemented in ℓ_∞ . Equivalently, c_0 is not an injective Banach space.

The contradiction shows that no such separable-range extension S exists. In particular, the norm-preserving extension demanded by the ℓ_∞ -extension property cannot exist. Hence $\mathfrak{X}$ is an injective operator ideal failing the ℓ_∞ -extension property. $\square$

Verification Notes

The proof uses only standard Banach-space facts: Sobczyk’s theorem and Phillips’ non-complementability theorem for $c_0 \subset \ell_\infty$. The contradiction rules out any separable-range extension of the Alouglu embedding i_{c_0} , so it is stronger than a failure of norm preservation.

Novelty and Literature Check

Before promotion, the lightweight run indexes

`registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, `proof_gaps/index.tsv`

were searched for 1207.1947, the source title, injective operator ideals, and ℓ_∞ -extension terminology; no duplicate solution to this question was found. Local arXiv-source search for the exact phrase “injective operator ideal that fails the ℓ_∞ -extension property” found the source paper. Web searches on June 28, 2026 for exact variants of “ ℓ_∞ -extension property” and “injective operator ideal” surfaced the source paper but no later explicit answer. A related paper, arXiv:1106.5088, records separable-range operators among classical injective and surjective ideals in an extension/lifting context, but it does not state this Karn–Sinha counterexample.

Human Review Recommendation

Review as a likely valid counterexample. The main points to check are: (1) the separable-range class is accepted as a Banach operator ideal with the operator norm; (2) Sobczyk’s theorem applies to the separable closed range E ; and (3) complementability in the Alouglu envelope indeed implies injectivity, hence contradicts Phillips’ theorem for c_0 .

References

- [1] A. K. Karn and D. P. Sinha, *Compactness and an approximation property related to an operator ideal*, arXiv:1207.1947, 2012.
- [2] A. Sobczyk, *Projection of the space (m) on its subspace (c_0)* , Bulletin of the American Mathematical Society 47 (1941), 938–947.
- [3] R. S. Phillips, *On linear transformations*, Transactions of the American Mathematical Society 48 (1940), 516–541.